

Riesz-energy, finiteness and every-density corollaries of planar universal optimality, kernel-checked in Lean 4

Xavier Callens (SocrateAI Lab)

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Abstract

The openai/math release contains an AI-generated Lean 4 proof that the triangular lattice A of covolume one minimises the lower g -energy among all locally finite planar configurations of density one, for every smooth completely monotone g (theorem `universal_energy_minimum`, Comparator-accepted by us under the pinned toolchain) [1, 2]. We derive three named consequences inside that pinned environment and check them with the Lean kernel, using only the axioms `propext`, `Classical.choice`, `Quot.sound`: (D1) for every $s > 0$ the Riesz potential $|x - y|^{-s}$ is admissible, so the triangular lattice minimises the Riesz s -energy among all locally finite density-one configurations; (D1b) for $s > 2$ the triangular lattice’s Riesz s -energy (an Epstein zeta value) is finite, so D1 is not the empty statement $\infty \leq \infty$ (for $0 < s \leq 2$ it is); (D2) for every density $\rho > 0$ and every admissible g , the rescaled lattice $\rho^{-1/2}A$ has density ρ and minimises the g -energy among all locally finite density- ρ configurations; without the rescaling the statement is false, which we also prove. Mathematically each step is short: complete monotonicity of $t^{-s/2}$ and a scaling argument. The upstream manuscript already states informally that its theorem covers inverse powers. What is new is the kernel-checked form of these statements, with a statement-fidelity audit, planted negative controls and independent re-verification. Everything is conditional on the upstream theorem, which has not been peer reviewed and which, by its authors’ own account, does not establish the stronger per-potential form of the Cohn–Kumar conjecture.

1 Context and scope

Rankin, Cassels, Ennola and Diananda [5, 6, 7, 8, 9] showed that the triangular lattice minimises the Epstein zeta function $\sum_{a \neq 0} |a|^{-s}$, $s > 2$, among planar lattices of fixed covolume, and Montgomery [10] did the same for Gaussian (theta) sums. Comparison with *all* configurations is the planar case of the universal-optimality question of Cohn and Kumar [3, Conj. 9.4], proved in dimensions 8 and 24 by Cohn, Kumar, Miller, Radchenko and Viazovska [4]. Partial planar results concern restricted competitor classes [11] and small perturbations [12].

On 2026-09-23 the openai/math release posted a manuscript [1] claiming the energy-comparison conclusion in the plane, together with a Lean proof of it. By its own account that manuscript does not construct a sharp auxiliary function for each potential, which is the stronger per-potential form of the Cohn–Kumar conjecture. The release of 2026-10-07 withdrew three unrelated manuscripts (Hodge/K3). It did not touch the triangular manuscripts, and the Lean files we use are byte-identical at its commit (Section 2).

What this note is and is not. It is a kernel-checked derivation, inside upstream’s pinned environment, of three statements physicists use: Riesz energies, their finiteness, and arbitrary density.

It is *not* an independent proof of universal optimality, *not* a new mathematical idea (the manuscript already notes that inverse powers are included), and *not* a peer review of the upstream proof. If the upstream statement is a faithful formalisation and its proof is correct, then D1–D2 are theorems; otherwise they are only implications.

2 The upstream statement and a fidelity audit

Let $\text{Plane} = \mathbb{R}^2$ with the Euclidean norm, and let $N_R(C)$ be the number of points of C in the closed disk of radius R about 0. The upstream Lean file (`ComparatorChallenges/TriangularEnergy.lean`, sha256 `af51979c...0616`) defines:

- g is *admissible* if $g \in C^\infty(0, \infty)$, $g \geq 0$, and $(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 0$ and $t > 0$;
- C has *density one* if $N_R(C)/(\pi R^2) \rightarrow 1$;
- $E_g(C) = \liminf_{R \rightarrow \infty} N_R^{-1} \sum_x \sum_{y \neq x} g(|x - y|^2)$, the sums running over points of C in the closed R -disk, with values in $[0, \infty]$;
- $A = \{b^{-1/2}(j + k/2, kb) : j, k \in \mathbb{Z}\}$, $b = \sqrt{3}/2$ (covolume one), and $E_g^{\text{lat}} = \sum_{a \in A \setminus 0} g(|a|^2) \in [0, \infty]$.

The theorem is: for admissible g and locally finite C of density one, $E_g^{\text{lat}} \leq E_g(C)$ and $E_g^{\text{lat}} = E_g(A)$.

We compared each item with the lower-energy and centred-density formulation of [4, Defs. 1.1–1.3] that the manuscript cites (`results/discovery/statement_fidelity.md`). They match, including the ordered-pair normalisation by N_R and the convention that divergent sums equal ∞ . Two points remain. First, upstream uses closed disks and we did not analyse whether the open-disk lower energy can differ. Second, the audit was done by the producing agent; the independent verifiers read the same definitions separately.

3 Results

All statements below are Lean theorems in the AutoevolveAI repository (`formal_cert/Triangular/`), compiled against upstream’s pinned Lean 4.34.1 / Mathlib `d13f23b7` / openai/math build. Here $\text{riesz}_s(t) = t^{-s/2}$, so $\text{riesz}_s(|x - y|^2) = |x - y|^{-s}$; the verifier kernel-checked this identity for $r > 0$.

Theorem 1 (D1, `TriangularRiesz.riesz_admissible`, `triangular_riesz_optimal`). *For every $s > 0$, riesz_s is admissible. Hence for every locally finite C of density one, $\sum_{a \in A \setminus 0} |a|^{-s} \leq E_{\text{riesz}_s}(C)$.*

Proof idea. Mathlib’s `Real.iter_deriv_rpow_const` gives $\frac{d^r}{dt^r} t^p = \text{desc}_r(p) t^{p-r}$. With $p = -s/2$, $\text{desc}_r(-s/2) = (-1)^r \text{asc}_r(s/2)$ (`ascPochhammer_eval_neg_eq_descPochhammer`), and $\text{asc}_r(s/2) > 0$, which gives the sign pattern. Smoothness away from 0 is `Real.contDiffAt_rpow_const_of_ne`. No value at $t \leq 0$ is ever used.

Theorem 2 (D1b, `TriangularRiesz.latticeEnergy_riesz_lt_top`). *For every $s > 2$, $\sum_{a \in A \setminus 0} |a|^{-s} < \infty$.*

Proof idea. `norm_sq_triangularPoint`: $|p(j, k)|^2 = \frac{2}{\sqrt{3}}(j^2 + jk + k^2)$. Reindex the sum over $A \setminus 0$ by the injective parametrisation, map $k \mapsto -k$ to match the form $j^2 - jk + k^2$, and apply the openai/math lemma `LatticeSummability.summable_normForm_neg_rpow` with $\sigma = s/2 > 1$. For

$0 < s \leq 2$ the sum diverges (partial sums grow like $2\pi \log R$ at $s = 2$; the verifier checked this numerically), so D1 is informative exactly for $s > 2$.

For $\rho > 0$ say that C has *density* ρ if $N_R(C)/(\pi R^2) \rightarrow \rho$. At $\rho = 1$ this is upstream’s definition by `Iff.rfl`.

Theorem 3 (D2, `TriangularDensity.universal_any_density`, `triangular_optimal_any_density`). *Let g be admissible and $\rho > 0$. Then $\rho^{-1/2}A$ is locally finite and has density ρ , and for every locally finite C of density ρ ,*

$$E_g(\rho^{-1/2}A) = \sum_{a \in A \setminus 0} g(|a|^2/\rho) \leq E_g(C).$$

Proof idea. $t \mapsto g(ct)$ is admissible for $c > 0$; the identity $\frac{d^r}{dt^r} g(ct) = c^r g^{(r)}(ct)$ holds globally from `deriv_comp_mul_left` with no differentiability hypothesis. Next, $N_R(\lambda C) = N_{R/\lambda}(C)$, so $\sqrt{\rho}C$ has density one, and the disk energies transform as $E_{g,R}(\lambda C) = E_{g(\lambda^2), R/\lambda}(C)$. The $\liminf$ is invariant under $R \mapsto R/\lambda$, because this order isomorphism maps `atTop` to itself. Applying upstream’s theorem to $g(\cdot/\rho)$ and $\sqrt{\rho}C$ gives the inequality, and its attainment conjunct gives the equality. The density and attainment lemmas (`scaled_density`, `attained`) were written by the independent D2 verifier and merged with attribution.

Corollary 4 (`riesz_any_density`). *For $s > 0$, $\rho > 0$ and locally finite C of density ρ : $\rho^{s/2} \sum_{a \in A \setminus 0} |a|^{-s} \leq E_{\text{riesz}_s}(C)$ (stated in Lean as $\sum g(\rho^{-1}|a|^2)$; informative for $s > 2$).*

Negative control (`TriangularDensity.unscaled_false`). The unrescaled claim “ $E_g^{\text{lat}} \leq E_g(C)$ ” for every locally finite C of density ρ is false at $\rho = 1/4$, $s = 4$, and this is proved in Lean. The witness $C = 2A$ has energy $\sum (4|a|^2)^{-2}$, which is strictly smaller, and finite by D1b. So the ρ^{-1} in D2 cannot be dropped. At $\rho > 1$ the unrescaled claim is true but weaker, because admissible g are non-increasing.

4 Verification

- **Kernel and axioms.** `Riesz.lean`, `RieszFinite.lean` and `AnyDensity.lean` compile with exit code 0 (each about 14–20s once upstream’s `oleans` are cached). `#print axioms` reports `{propext, Classical.choice, Quot.sound}` for every theorem above and for upstream’s `universal_energy_minimum`. A planted `sorry` and a planted axiom in the same audit file are both flagged (`sorryAx`, `ctl_cheat`). There is no `sorry` or `native_decide` in the shipped files.
- **Preregistration.** D1b and D2 were preregistered with kill criteria before their Lean files were written (`results/discovery/D1b,D2/preregistration.json`). D1 was preregistered only through the proposal document (`docs/DISCOVERY_PROPOSAL_2026-10-10.md`), which assigned it prior probability about 85%.
- **Independent verification.** D1/D1b and D2 were each re-verified by a separate AI-agent instance that recompiled the files and wrote its own axiom audit, controls and numerics. Both verdicts were `CONFIRMED_WITH_CORRECTIONS`, and all corrections were wording or scope, now applied. The verifiers’ extra controls included a kernel proof that `riesz-2` ($t \mapsto t$) is *not* admissible, and a check that the D1b proof cannot pass its only s -dependent step at $s = 2$.

- **Numerics** (verifier code, mpmath; closed forms agree with direct sums to about 10^{-7}). Table 1: the triangular lattice beats the square lattice at equal density in every case, and for Riesz $s = 4$ the ratio 0.959605 does not depend on ρ , as homogeneity predicts.

$g(t)$	ρ	triangular	square	square – triangular
$t^{-3/2}$ ($s = 3$)	1	8.8927451004	9.0336216831	0.1409
t^{-2} ($s = 4$)	1	5.7833592997	6.0268120397	0.2435
t^{-3} ($s = 6$)	1	4.1412565472	4.6589136156	0.5177
e^{-t}	0.5	0.60238788592	0.61630926604	1.39×10^{-2}
e^{-t}	2	5.2831853119385	5.2831853744169	6.25×10^{-8}
e^{-t}	4	11.566370614359172955	11.566370614359173314	3.59×10^{-16}

Table 1: Per-point lattice energies $\sum_{y \in L \setminus 0} g(|y|^2)$ at density ρ (independent verifier numerics). At $\rho = 4$ the Gaussian gap is below double precision, and a float64 run misreports the order; the 60-digit run resolves it.

5 Limitations

1. Everything is conditional on upstream’s `universal_energy_minimum`. That proof is AI-generated and has not been peer reviewed, though we checked that it is accepted by Comparator, with clean axioms, under the pinned toolchain. Its statement has been audited by AI agents, not by a human expert.
2. The Cohn–Kumar conjecture in its per-potential form (a sharp auxiliary function for each potential) is not established, upstream or here.
3. For $0 < s \leq 2$, D1 only says $E(C) = \infty$; renormalised Riesz and Coulomb energies are treated in the upstream manuscript, not here.
4. Closed versus open disks in the energy definition was not analysed.
5. Other admissible families (Yukawa, $e^{-\kappa\sqrt{t}}/\sqrt{t}$) were not attempted. They need complete monotonicity of compositions, which current Mathlib does not offer in usable form.

Reproducibility. Lean sources: `Riesz.lean`, `RieszFinite.lean`, `AnyDensity.lean` and `Axioms.lean` in `formal_cert/Triangular/` (sha256 in the bundle manifest). Upstream: openai/math commit `adc7f124`, unchanged at `fd4aeeb2` for the files used. Toolchain: Lean 4.34.1, Mathlib `d13f23b7`. Compiler driver: `scripts/certified_numerics/lean_pinned.py`. Verification reports and numerics scripts: `results/discovery/`. Repository: <https://github.com/xaviercallens/AutoevolveAI>, branch `worktree-openai-math-discovery`. Method: SocrateAI-Scientific-Elenchus (tiered hypotheses with predictions and kill criteria) [13].

AI use. The Lean proofs, audit and text were produced by AI agents (Claude) under the author’s direction; the verification runs were separate agent instances. The author takes responsibility for the claims as scoped above.

References

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